Green Algorithms calculator

What's the carbon footprint of your computations?

Classic view

AI view

Change language

ⓘ Change data version

Some news...

⚡ We are now using ElectricityMap's data for carbon intensity of electricity usage!

🌱 v3.1 out: you can now input your own carbon footprint, the underlying data is now in a different GitHub repo for better reproducibility, and we're ready for multi-language support. This prepares the upcoming major update to v4.0, stay tuned!

🌍 Multi-language is enabled, we just need to add the languages! If you want to contribute, you can submit a Pull Requst directly on the GitHub repository. All you have to do is to edit this file by adding your own language's translation to each key, we'll do the rest!

🐞 It's always possible that some bugs have slipped through the net of this new release... If you spot one, just let us know here.

More on the project website!

Details about your algorithm

To understand how each parameter impacts your environmental footprint, check out the formula below and the methods article

Runtime (HH:MM)

504 | 0

Type of cores

CPU ⓘ

Number of cores used

12 ⓘ

Model

I can't find my CPU

If your CPU is not in the list, you can manually input the few key specs used by the calculator. Pre-filled values are average ones.

Number of cores

12

TDP (in Watt)

120

Die area (in cm2)

12

Memory available (in GB)

10

Select the platform used for the computations

Local server

Select location

World

Carbon intensity : **480** gCO_2eq/kWh — source

Do you know the real usage factor of your CPU?

(•) Yes ( ) No

1

Do you know the Power Usage Efficiency (PUE) of your local data centre?

(•) Yes ( ) No

1,67

Do you want to use a multiplicative factor?

( ) Yes (•) No

Share your results as a csv file!

Import results

Drag and drop or click to select your .csv file.

49.99 kgCO2e

Carbon footprint

104.14 kWh

Energy needed

4.54 tree-years

Carbon sequestration

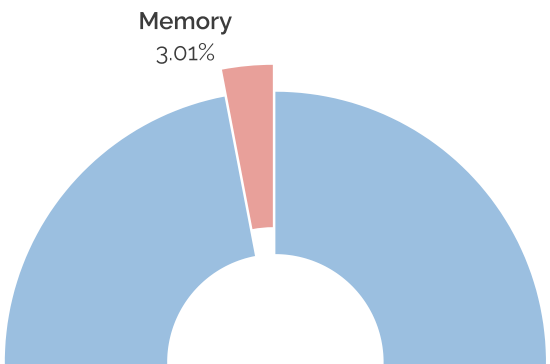
285.63 km

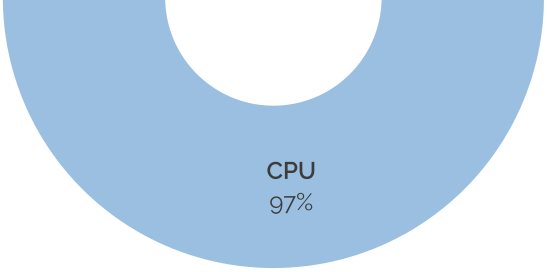
in a passenger car ⓘ

45%

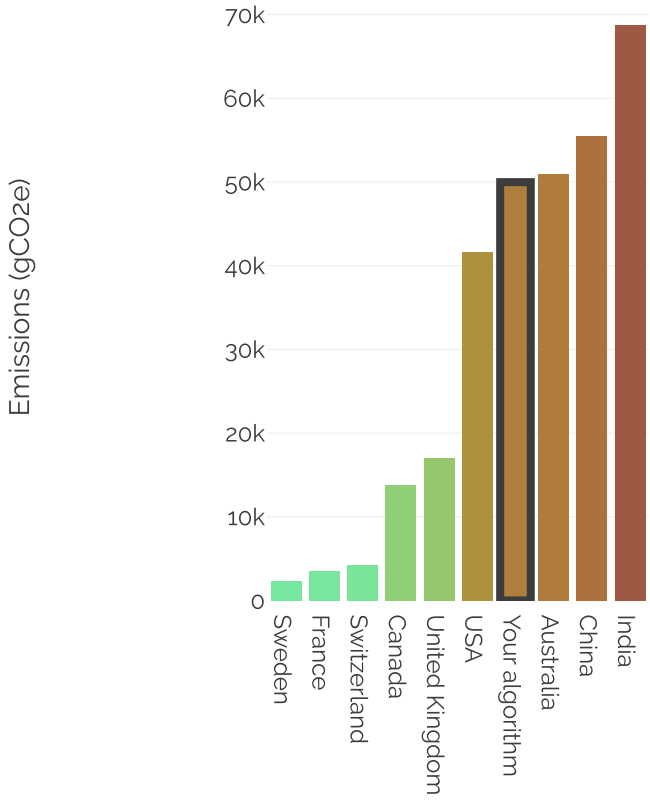
of a flight Paris-Dublin

Computing cores VS Memory





How the location impacts your footprint



More details about the methodology in the methods paper.

Other resources you may find interesting on this topic: the GREENER principles for environmentally sustainable computational science, or this short primer discussing different options for carbon footprint estimation.

Using a SLURM-powered HPC server? Check out GA4HPC and the recent Green Algorithms Dashboard, they are great way to visualise the carbon footprint of all your jobs at once!

The formula

The carbon footprint is calculated by estimating the energy draw of the algorithm and the carbon intensity of producing this energy at a given location:

```
carbon footprint = energy needed * carbon intensity
```

Where the energy needed is:

```
runtime * (cores power draw * usage + memory power draw) * PUE * multiplicative factor
```

The power draw of the computing cores depends on the model and number of cores, while the memory power draw only depends on the size of memory *available*. The usage factor corrects for the real core usage (default is 1, i.e. full usage). The PUE (Power Usage Effectiveness) measures how much extra energy is needed to operate the data centre (cooling, lighting etc.). The multiplicative factor is used to take into account multiple identical runs (e.g. for testing or optimisation).

The Carbon Intensity depends on the location and the technologies used to produce electricity. If you want to check out the carbon intensity in real time, and see discrepancies between countries, check out the ElectricityMap website. Also, note that **the "energy needed" indicated at the top of this page is independent of the location.**

About CO2e

"Carbon dioxide equivalent" (CO2e) measures the global warming potential of a mixture of greenhouse gases. **It represents the quantity of CO2 that would have the same impact on global warming** as the mix of interest and is used as a standardised unit to assess the environmental impact of human activities.

What is a tree-month?

It's the amount of CO2 sequestered by a tree in a month. **We use it to measure how long it would take to a mature tree to absorb the CO2 emitted by an algorithm.** We use the value of 11 kg CO2/year, which is roughly 1kg CO2/month.

How to report it?

It's important to track the impact of computational research on climate change in order to stimulate greener algorithms. For that, **we believe that the carbon footprint of a project should be reported on publications alongside other performance metrics**.

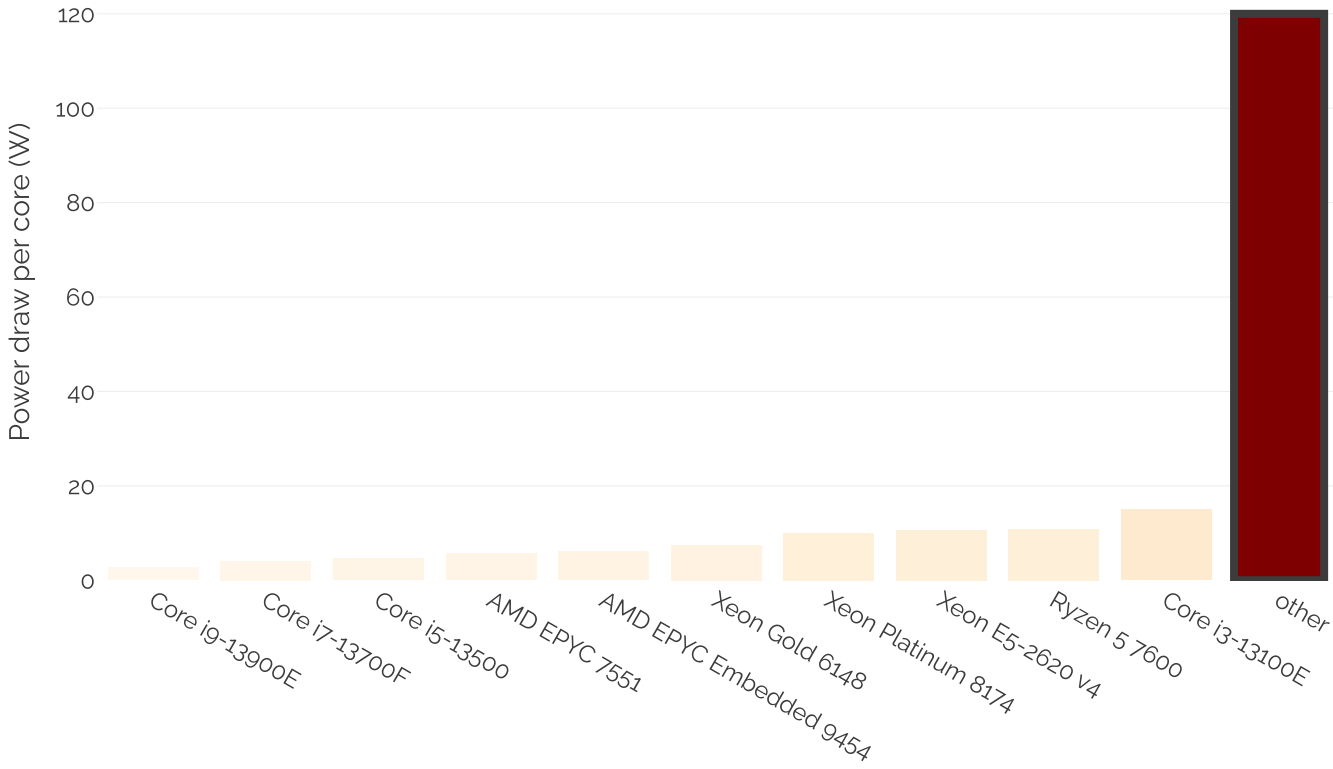
Here is a text you can include in your paper:

> *This algorithm runs in 504h on 12 CPUs other, with 10 GB of allocated memory. It draws 104.14 kWh. Based in Global, this has a carbon footprint of 49.99 kgCO2e, which is equivalent to 4.54 tree-years (calculated using green-algorithms.org v3.1 [1]).*

[1] Lannelongue, L., Grealey, J., Inouye, M., Green Algorithms: Quantifying the Carbon Footprint of Computation. Adv. Sci. 2021, 2100707.

Including the version of the tool is useful to keep track of the version of the data used.

Power draw of different processors



Data and code

All the data and code used to run this calculator can be found on GitHub

Questions / Suggestions?

If you have questions or suggestions about the tool, you can open an issue on the Github.

How to cite this work

Lannelongue, L., Grealey, J., Inouye, M., Green Algorithms: Quantifying the Carbon Footprint of Computation. Adv. Sci. 2021, 2100707 https://doi.org/10.1002/advs.202100707.

About us

The Green Algorithms Initiative is led by the Cambridge Sustainable Computing Lab at the University of Cambridge, but made possible by the contribution and support of many: full list here.

In particular, we are thankful for the support of the Wellcome Trust, NIHR Cambridge Biomedical Research Centre, and French Department for the Ecological Transition.

#ShowYourStripes

These coloured stripes in the background represent the change in world temperatures from 1850 to 2018. This striking design was made by Ed Hawkins from the University of Reading.

More on ShowYourStripes.info

Additional credits for the app can be found on the GitHub